

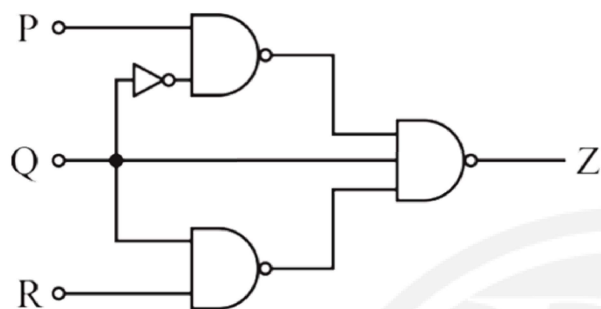
CS & IT

Digital Logic

Optimisation

DPP: 1

Q1 For a 3-input logic circuit shown below, the output Z can be expressed as

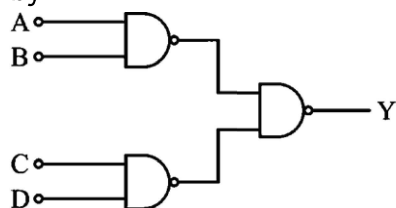


- (A) $Q + \bar{R}$ (B) $P\bar{Q} + R$
 (C) $\bar{Q} + R$ (D) $P + \bar{Q} + R$

Q2 The complete set of only those Logic Gates designated as Universal Gates is

- (A) NOT, OR and AND Gates
 (B) XNOR, NOR and NAND Gates
 (C) NOR and NAND Gates
 (D) XOR, NOR and NAND Gates

Q3 In the logic circuit shown in the figure, Y is given by



- (A) $Y = ABCD$
 (B) $Y = (A + B)(C + D)$
 (C) $Y = A + B + C + D$
 (D) $Y = AB + CD$

Q4 $F = AB + CD + E$ will be implemented with how many minimum number NAND gates?

- (A) Three (B) Four

(C) Five

(D) Six

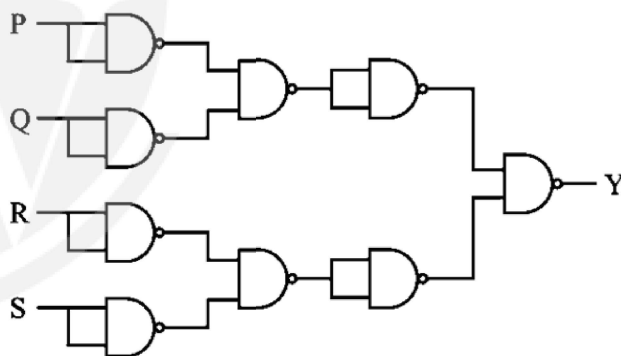
Q5 The minimum number of NAND gates required to reduce the expression $((A + B)C)D$ is

- (A) 6 (B) 5
 (C) 8 (D) 4

Q6 In a two-input NAND gate, if both inputs are shorted, it will behave like a _____ gate.

- (A) Buffer (B) AND
 (C) NOT (D) EX-OR

Q7 For the circuit shown in figure the Boolean expression for the output Y in terms of inputs P, Q, R and S is



- (A) $\bar{P} + \bar{Q} + \bar{R} + \bar{S}$
 (B) $P + Q + R + S$
 (C) $(\bar{P} + \bar{Q}) + (\bar{R} + \bar{S})$
 (D) $(P + Q)(R + S)$

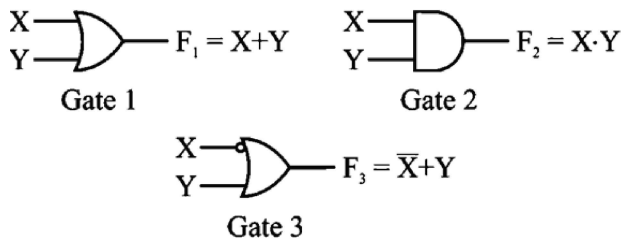
Q8 A universal logic gate can implement any Boolean function by connecting sufficient number of them appropriately. Three gates are shown:



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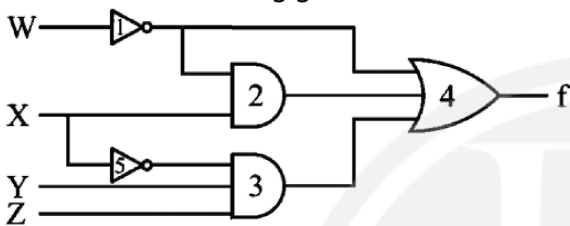
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Which one of the following statements is TRUE ?

- (A) Gate 1 is a universal gate.
- (B) Gate 2 is a universal gate.
- (C) Gate 3 is a universal gate
- (D) None of the shown is a universal gate.

Q9 Consider the following gate network:



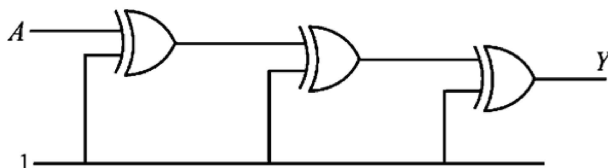
Which one of the following gates is redundant?

- (A) Gate No. 1
- (B) Gate No. 2
- (C) Gate No. 3
- (D) Gate No. 4

Q10 The minimum of NAND gates required to implement $A + A B C$ is equal to

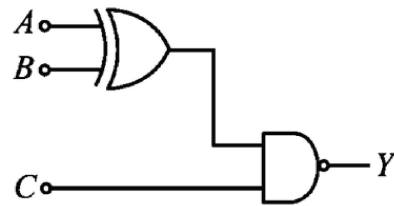
- (A) 0
- (B) 1
- (C) 4
- (D) 7

Q11 The initial output of the following circuit is 1. If we apply 010101 at input A (first bit is zero), then what is the bit pattern generated at the output Y.



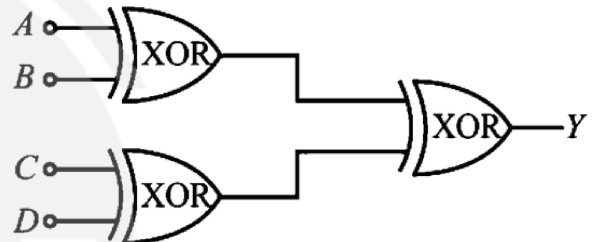
- (A) 010101
- (B) 101010
- (C) remains at 0
- (D) remains at '1'

Q12 The Boolean expression of the output of the logic circuit shown in figure is



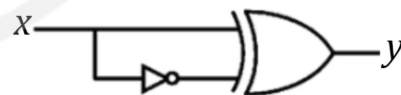
- (A) $Y = \bar{A} \bar{B} + AB + \bar{C}$
- (B) $Y = \bar{A} \bar{B} + AB + C$
- (C) $Y = \bar{A} B + \bar{A} \bar{B} + C$
- (D) $Y = \bar{A} B + \bar{A} \bar{B} + C$

Q13 A, B, C and D are input, and Y is the output bit in the XOR gate circuit of the figure below. Which of the following statements about the sum S of A, B, C, D and Y is correct?



- (A) S is always either zero or odd.
- (B) S is always either zero or even.
- (C) $S = 1$ only if the sum of A, B, C and D is even.
- (D) $S = 1$ only if the sum of A, B, C and D is odd.

Q14 The output Y of the logic circuit given below is



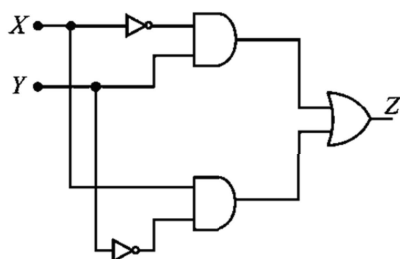
- (A) 1
- (B) 0
- (C) x
- (D) $\bar{x}$

Q15 A bulb in a staircase has two switches, one switch being at the ground floor and the other one at the first floor. The bulb can be turned ON and also can be turned OFF by any one of the switches irrespective of the state of the other switch. The logic of switching of the bulb resembles



- (A) An AND gate
 (B) An OR gate
 (C) An XOR gate
 (D) A NAND gate

- Q16** In the circuit shown below, X and Y are digital inputs, and Z is a digital output. The equivalent circuit is



- (A) XNOR gate (B) NOR gate
 (C) NAND gate (D) XOR gate
- Q17** Which one of the following gate is also known as equivalence gate?
 (A) EX-OR (B) NAND
 (C) EX-NOR (D) NOR
- Q18** The logic function $f = \overline{xy} + \overline{x} \overline{y}$ is equal to
 (A) EX-NOR (B) NAND
 (C) EX-OR (D) NOR
- Q19** The minimum number of 2 input NOR gates required to implement a 2 input XOR gate is_____.
- Q20** The minimum number of 2 input NAND gates required to implement a 2 input EX-NOR gates is
 (A) 4 (B) 5
 (C) 3 (D) 6
- Q21** Find out minimize expression for following function
 $f = (A + B + C)(A + B + \overline{C})$
 (A) $(A + C)B$ (B) $A + B$
 (C) $AB\overline{C}$ (D) $A(B + C)$

- Q22** The Boolean expression

$$f = (X + Y)(X + \overline{Y}) + \overline{X} \overline{Y} + \overline{X}$$

- (A) Y (B) X
 (C) $\overline{X} \overline{Y}$ (D) $X + \overline{Y}$

- Q23** The logic expression

$$f = X + \overline{X}Y$$

Is equivalent to

- (A) $X + Y$ (B) XY
 (C) $\overline{X} + Y$ (D) $X + \overline{Y}$

- Q24** The logic expression

$$f = (A + B)(A + C)$$

Is equivalent to

- (A) $A + BC$ (B) $B + AC$
 (C) $C + AB$ (D) $\overline{A} + BC$

- Q25** The Boolean expression

$$f = (1 + \overline{A})(B + AC)$$

is equivalent to

- (A) $AC + B$ (B) $\overline{A}C + B$
 (C) $\overline{A} + BC$ (D) 1

- Q26** Find minimization expression

$$f = (A + \overline{A}) + (BC + AC)(A + D)$$

- (A) 1 (B) 0
 (C) $AB + CD$ (D) $ABC + BCD + ACD$

- Q27** Find out minimization

$$f = (A + B)(A + B + C)$$

- (A) $B + C$ (B) $A + B$
 (C) $A + B + C$ (D) $AB + BC + AC$



Answer Key

Q1 (C)
Q2 (C)
Q3 (D)
Q4 (D)
Q5 (B)
Q6 (C)
Q7 (B)
Q8 (C)
Q9 (B)
Q10 (A)
Q11 (B)
Q12 (A)
Q13 (B)
Q14 (A)

Q15 (C)
Q16 (D)
Q17 (C)
Q18 (C)
Q19 5~5
Q20 (B)
Q21 (B)
Q22 (B)
Q23 (A)
Q24 (A)
Q25 (A)
Q26 (A)
Q27 (B)



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